**AMERICAN INTERNATIONAL UNIVERSITY–BANGLADESH (AIUB)**

**Faculty of Science and Technology**

**Dept. of Computer Science**

**COURSE: SOFTWARE ENGINEERING**

**Spring 2025-2026**

**Section: F**

**Group No :**
**Date: 21/04/2026**

**Software Requirement Specification [SRS] Document On**

**Project Name : SkillSwap**
**Supervised By: Mejbah Ahammad**

**Submitted By**

| Name | | ID |
|---|---|---|
| 1. | MD.LEON MIA | 23-53551-3 |
| 2. | MD. ABUBOKOR SIDDIK ROJAN | 22-47890-2 |
| 3. | MONIA ISLAM MUKTI | 24-55933-1 |
| 4. | FARHAN KABIR | 23-53686-3 |

# Table of Contents

1.INTRODUCTION .....3
2.SCOPES & FEATURES .....4
2.1.Scope 1 .....4
2.2.Scope 2 .....4
3.Scope 3 .....5
4.Scope 3 .....5
3.USE CASE DIAGRAM .....6
5.USER STORY TABLE .....7
6.REQUIREMENT TRACEABILITY MATRIX .....8
6.1.Functional Requirements .....8

# 1.INTRODUCTION

**SkillSwap** is a web-based peer-to-peer knowledge exchange platform designed to foster community-driven learning. The proposed platform allows users to act as both teachers and learners in a reciprocal environment, facilitating the exchange of diverse skills without the need for financial transactions. Through a centralized system, **SkillSwap** enables users to manage their skill profiles, discover compatible learning partners via an automated matching algorithm, initiate secure exchange requests, and build community reputation through a transparent rating and review system.

The main goal of this project is to democratize skill acquisition by removing financial barriers and providing a structured alternative to unorganized social media networking. By digitizing the exchange process, **SkillSwap** addresses issues such as the difficulty in finding verified learning partners, inconsistent communication, and a lack of accountability in informal skill sharing. This platform will enhance accessibility to new competencies, promote meaningful community collaboration, and provide a transparent, feedback-driven experience for every participant, effectively creating a "community currency" based on knowledge rather than capital.

# 2.SCOPES & FEATURES

## 2.1.Scope 1

This scope covers user identity management and skill classification. The system will allow users to register, create detailed profiles, and explicitly define their expertise ("Offer Skills") and learning objectives ("Wanted Skills"). By structured categorization, the platform builds a searchable knowledge base of the community's capabilities.

**Main features:**

- User registration and secure login.
- Comprehensive profile management (Bio, Profile Picture).
- Dynamic Skill-set definition (Adding/Editing Offer & Wanted skills).
- Privacy settings for personal visibility.

## 2.2.Scope 2

This scope covers the discovery ecosystem. The system will employ matching algorithms to connect users based on complementary skill sets—linking those who *offer* what another *wants*. It allows users to navigate the community effectively through categorization and filtering tools.

**Main features:**

- Smart Matching algorithm (Suggesting compatible partners).
- Advanced Search functionality (by skill category or location).
- Filtering options for refined discovery.
- Interactive user profiles for previewing potential exchange partners.

**3.Scope 3**

This scope covers the lifecycle of an exchange request. It manages the communication flow between users from the initial request proposal to the final agreement. The platform ensures user safety and data privacy by restricting contact information disclosure until a request is mutually accepted.

**Main features:**

- Initiating and sending exchange requests.
- Real-time notification system for requests and status updates.
- Request decision workflow (Accept/Reject/Pending).
- Secure contact information revelation (only upon successful acceptance).

**4.Scope 3**

This scope covers community trust and performance tracking. To maintain high-quality interactions, the platform will implement a feedback mechanism. Both parties can rate their experience and provide reviews upon completing an exchange, which contributes to their overall **Trust Score** and reputation within the platform.

**Main features:**

- Rating and Review system (1-5 star scale).
- Calculation and display of user **Trust Score**.
- Exchange history and task status tracking (In-progress/Completed).
- User dashboard for monitoring overall engagement and reputation.

## 3.USE CASE DIAGRAM

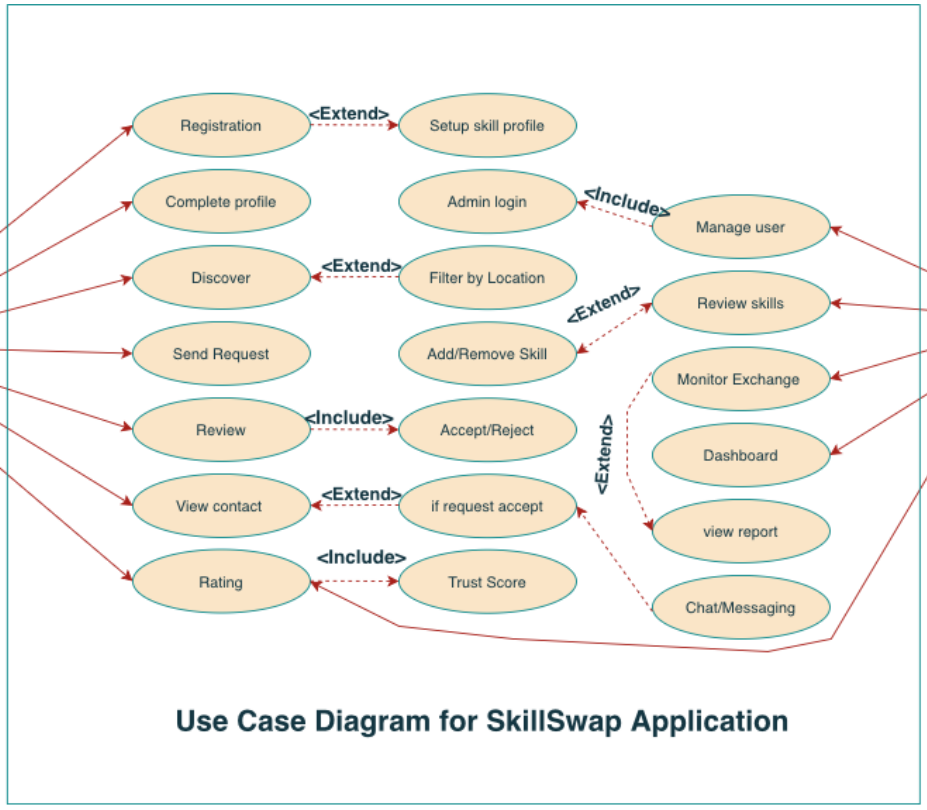
Use Case Diagram Description :

**Actors:**

- **User:** The core participant of the platform who dynamically acts as both a **Skill Provider** and a **Skill Requester**.
- **Admin:** The system supervisor responsible for oversight, user management, and platform analytics.

**Use Case Descriptions:**

- **User:**
  - **Account Actions:** Register, Login, Setup Skill Profile, Complete Profile.
  - **Discovery & Exchange:** Discover/Search Skills, Filter by Location, Send Swap Request, Accept/Reject incoming requests.
  - **Communication & Trust:** View Contact Details (WhatsApp/Email), Chat/ Messaging, Provide Ratings, Give Reviews.
- **Admin:**
  - **System Oversight:** Login, Manage Users (Ban/Approve), Monitor Exchange activity, View System Reports/Analytics, Dashboard access.



## 5.USER STORY TABLE

| User | User story |
|---|---|
| Admin | As an admin, I want to manage user accounts so that I can maintain community safety and verify authenticity. |
| Admin | As an admin, I want to moderate reports and flagged profiles so that the platform remains secure for all users. |
| **User** | As a user, I want to define my "Offer" and "Wanted" skills so that the system can match me with the right partners. |
| **User** | As a user, I want to search and filter for specific skills so that I can find a tutor who fits my learning goals. |
| **User** | As a user, I want to send an exchange request to another person so that we can initiate a learning session. |
| **User** | As a user, I want to accept or reject requests so that I can manage my availability and schedule effectively. |
| **User** | As a user, I want to view my partner's contact details only after the request is accepted so that my privacy is protected. |
| **User** | As a user, I want to provide a rating and review after an exchange so that I can help build community trust. |

# 6.REQUIREMENT TRACEABILITY MATRIX

## 6.1.Functional Requirements

The following section outlines the Requirement Traceability Matrix (RTM), which serves to map the project's functional and non-functional requirements to ensure alignment with the overall system objectives.

- The system shall allow users to register and manage their secure profiles.
- The system shall allow users to add and edit "Offer Skills" and "Wanted Skills."
- The system shall provide an automated matching feature based on user skill profiles.
- The system shall allow users to search and filter peers by skills or location.
- The system shall enable users to send, accept, and reject exchange requests.
- The system shall ensure contact information is hidden until a request is accepted.
- The system shall allow users to rate and review their partners after an exchange.
- The system shall display a trust score for users based on community feedback.

## 6.2. Non-Functional Requirements

- The system should be available 24/7 for seamless community interaction.
- The system should provide a modern, clean, and user-friendly interface.
- The system should respond to user queries and matching requests within a reasonable time.
- The system should protect user data and privacy through secure access controls.
- The system should be scalable to accommodate an increasing number of users.
- The system should maintain data accuracy and reliability throughout the exchange process.